

**FAKULTI TEKNOLOGI DAN KEJURUTERAAN ELEKTRONIK &
KEJURUTERAAN KOMPUTER**

ASSIGNMENT

BERR 2243 Database and Cloud System

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1. Introduction

The objective of this project is to design and implement a complete end-to-end IoT cloud system integrating an ESP32 microcontroller, EMQX MQTT broker, Node-RED middleware, MongoDB Atlas database, and MongoDB Charts dashboard. The system enables real-time collection, processing, storage, and visualization of environmental sensor data using a secure cloud architecture.

2. System Architecture Overview

The system consists of five main layers: Edge Layer (ESP32), Cloud Middleware (EMQX and Node-RED), Database Layer (MongoDB Atlas), Security Layer, and Application Layer (MongoDB Charts). Data is transmitted using MQTT protocol in JSON format from ESP32 to cloud services.

Edge Layer: ESP32 publishes temperature and humidity data via MQTT.

Cloud Layer: EMQX handles MQTT messaging; Node-RED processes and validates data.

Database Layer: MongoDB Atlas stores metadata and time-series sensor data.

Security Layer: RBAC and authentication restrict unauthorized access.

Application Layer: MongoDB Charts visualizes real-time and historical data.

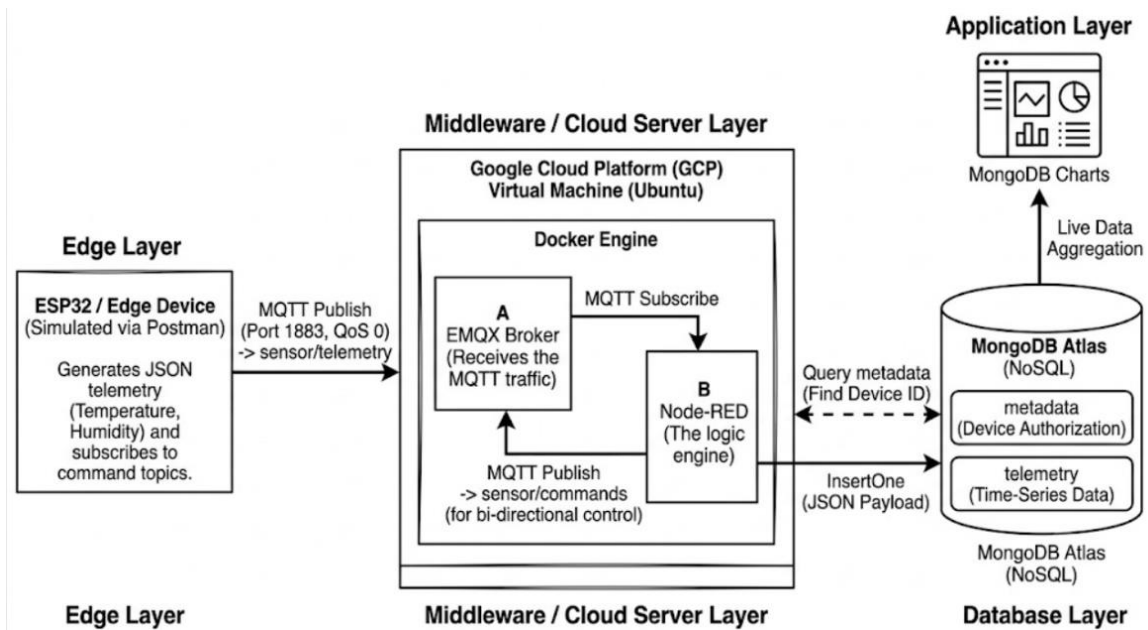
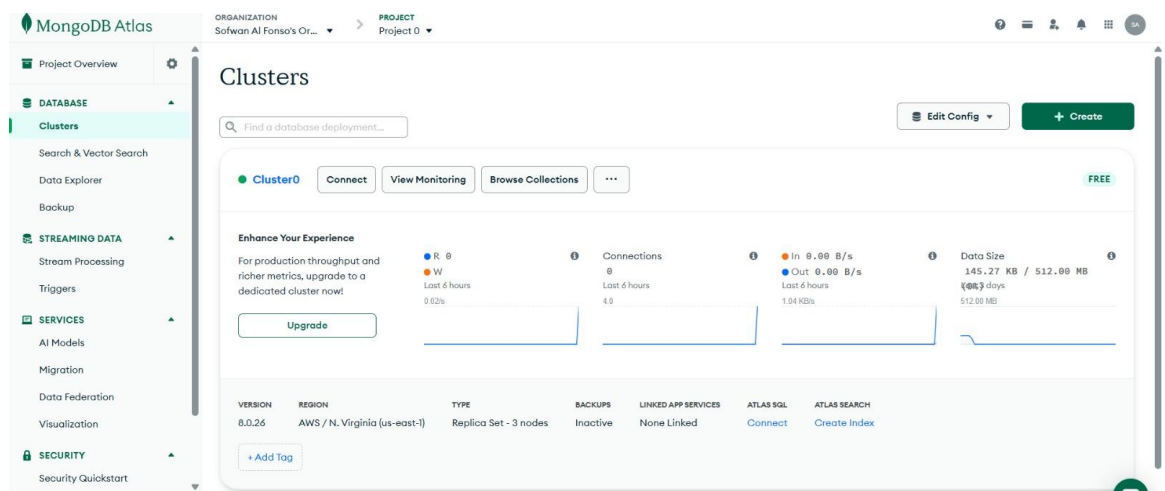


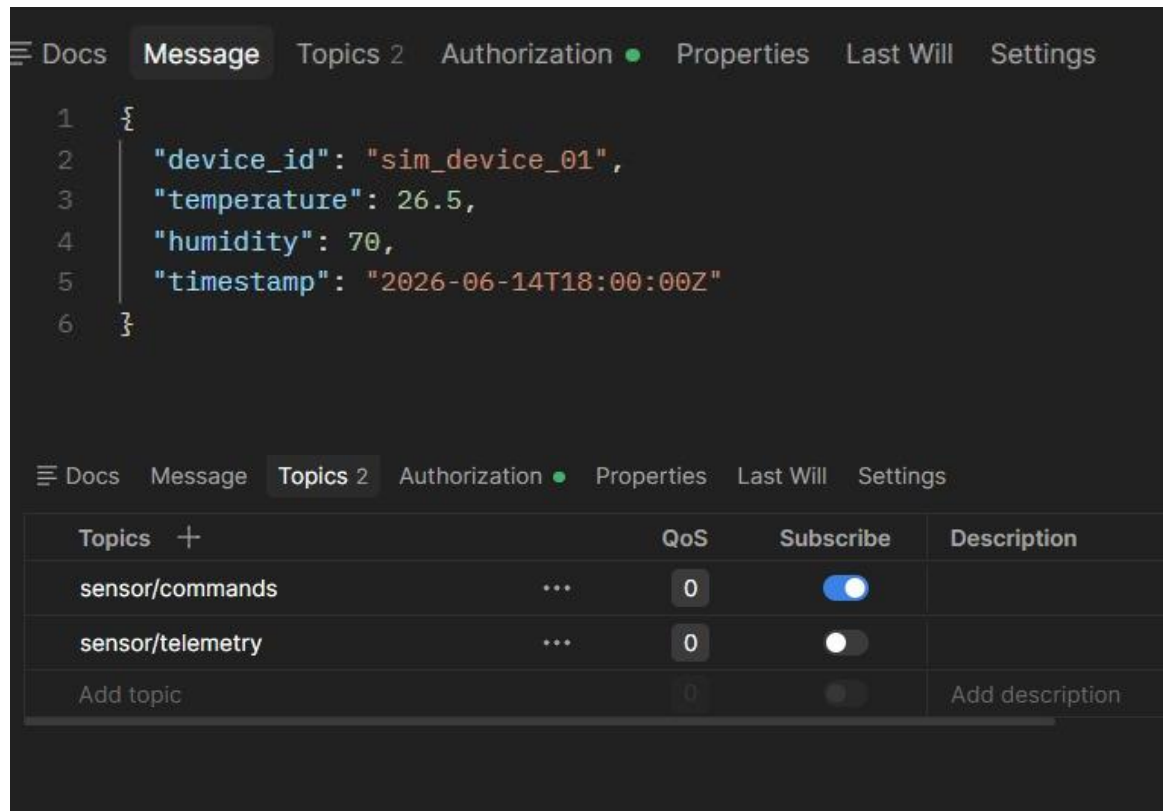
Figure 1: System Architecture

3. Deployment Process

3.1 Create cluster 0 at MongoDB Atlas



3.2 ESP32 programmed using postman to publish JSON sensor data.



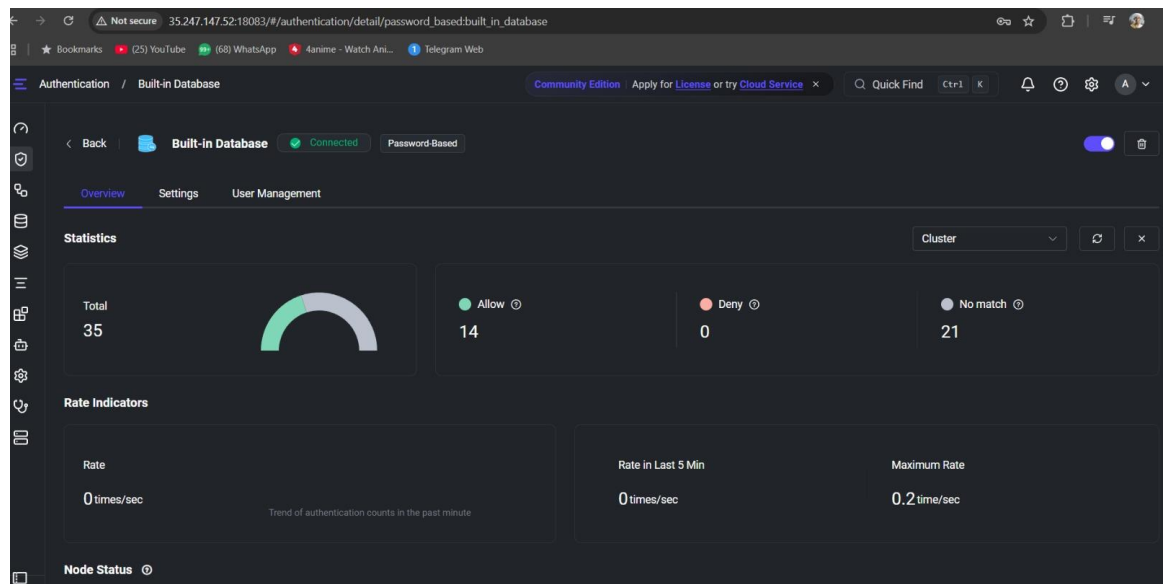
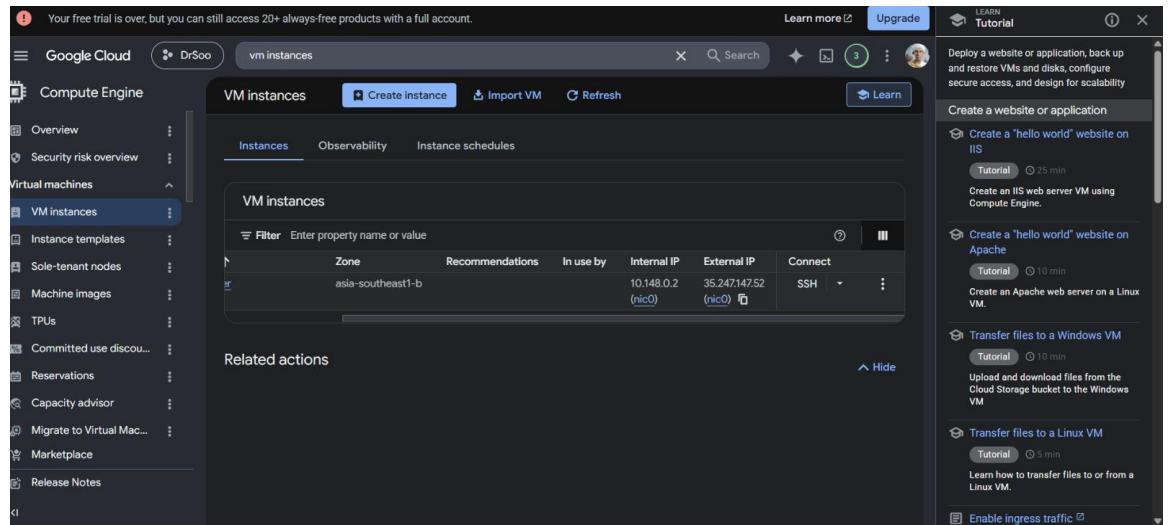
The screenshot displays the MQTT Explorer interface. The top section shows a message in JSON format:

```
1 {  
2   "device_id": "sim_device_01",  
3   "temperature": 26.5,  
4   "humidity": 70,  
5   "timestamp": "2026-06-14T18:00:00Z"  
6 }
```

The bottom section shows a table of topics:

Topics +	QoS	Subscribe	Description
sensor/commands ...	0	☒	
sensor/telemetry ...	0	☐	
Add topic	0	☐	Add description

3.3 Cloud VM deployed with EMQX and Node-RED using Google Cloud.

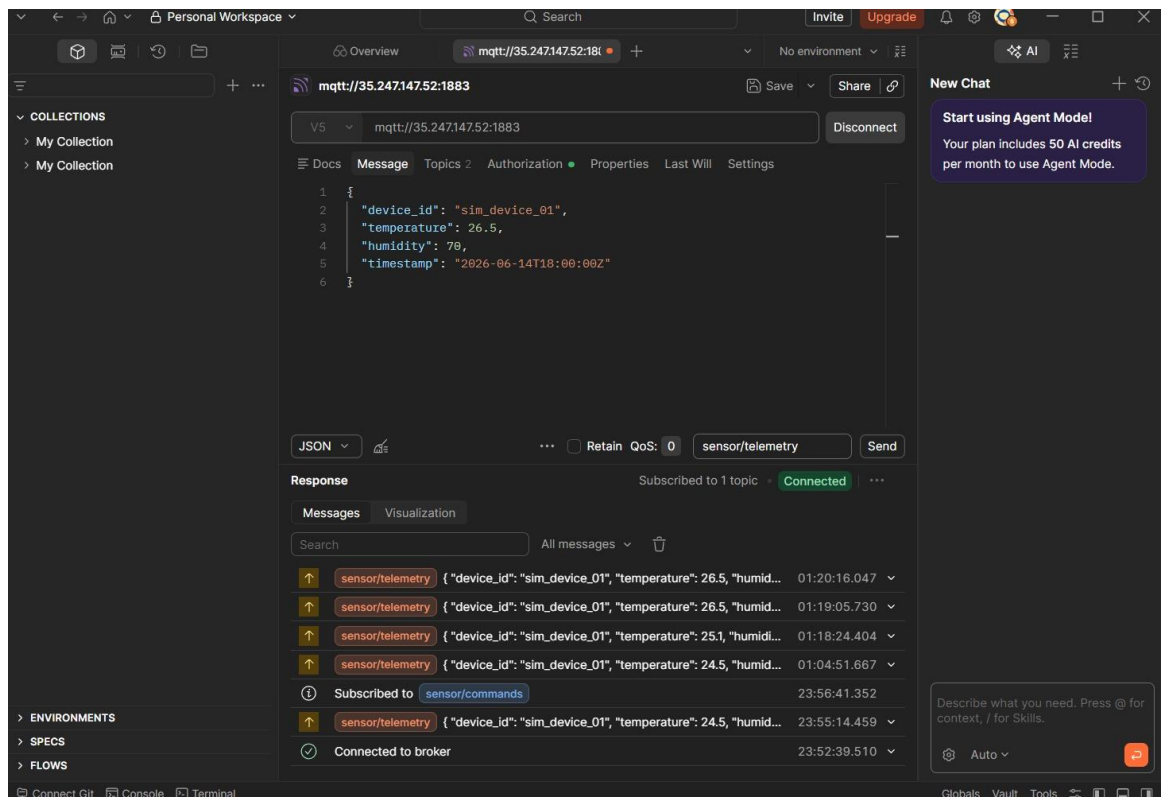


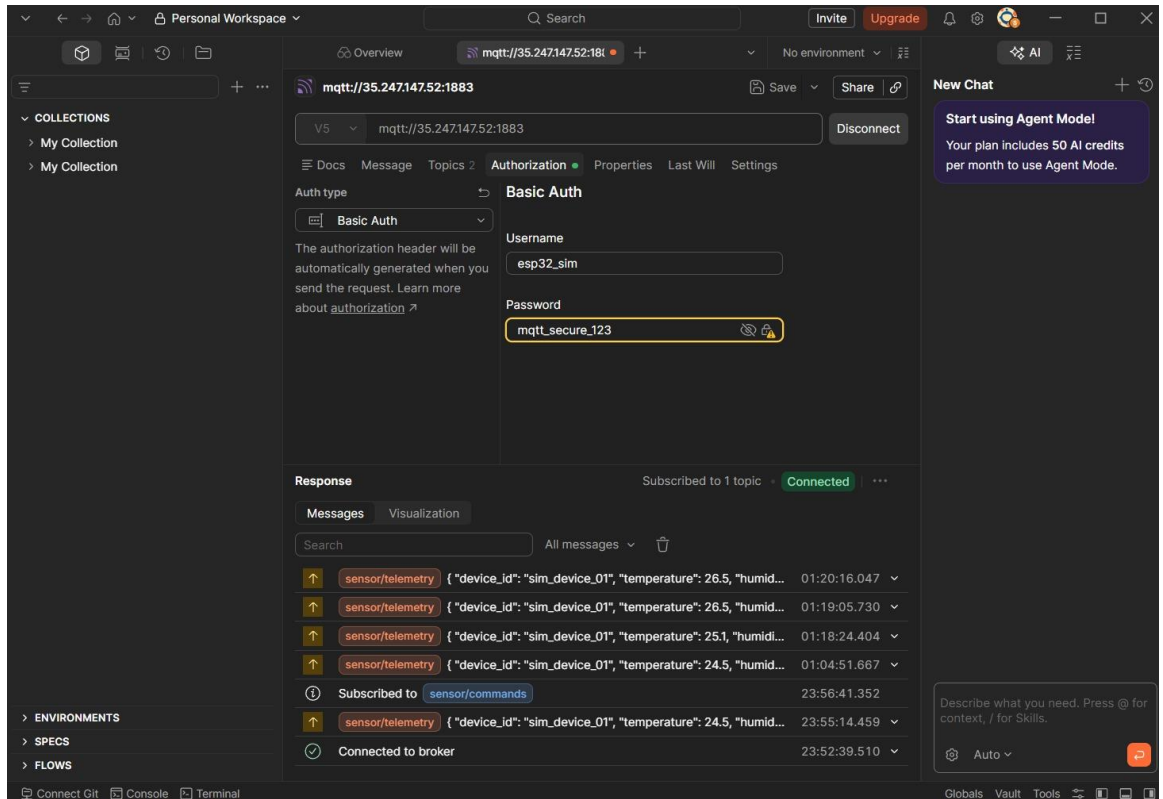
```
ssh.cloud.google.com/v2/ssh/projects/project-fb43babe-071b-46f5-ab5/zones/asia-southeast1-b/instances/iot-server?authuser=0&hl=en...
ssh.cloud.google.com/v2/ssh/projects/project-fb43babe-071b-46f5-ab5/zones/asia-southeast1-b/instances/iot-server?authuser=0...

SSH-in-browser
[Icons: UPLOAD FILE, DOWNLOAD FILE, Help, SSH Icon, Settings]

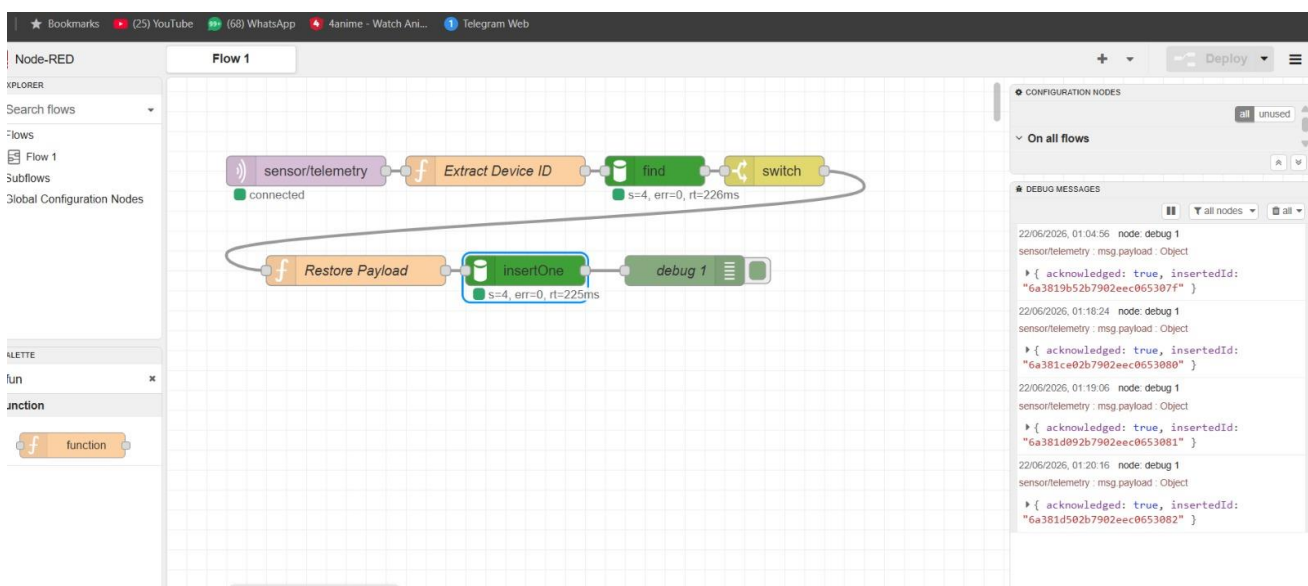
Digest: sha256:01cd0f37fe258998bea07ed25fafbbdb00eb35daf59fd1491f6f1b4881d49961
Status: Downloaded newer image for emqx/emqx:latest
38bc49be18a419904686d5b0781f5704caa619bbb983de29592730f626c8e40d
Unable to find image 'nodered/node-red:latest' locally
latest: Pulling from nodered/node-red
eea2c8c31119: Pull complete
6a0ac1617861: Pull complete
414f813165cf: Pull complete
8669bb364265: Pull complete
c7fd89fde416: Pull complete
9d3bbd0162a3: Pull complete
7486a1919852: Pull complete
f71d89736f68: Pull complete
4f4fb700ef54: Pull complete
4c67b0d34709: Pull complete
a182acd4b436: Pull complete
01f47b9bc472: Pull complete
f8c7aafc4dbd: Pull complete
d0c2857a6a41: Pull complete
b98901e02247: Pull complete
df375f62720a: Pull complete
0611226a663d: Pull complete
4f14fbca93ad: Download complete
Digest: sha256:153f411d2993abd9ccd8290017ff2bb531326320b5cd3b200c1a4bb1339eb819
Status: Downloaded newer image for nodered/node-red:latest
63513e2a48039620092bcbef553510498350f6094f385b18a8aa0bcd2e8605a5
sofwansabri@iot-server:~$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS         NAMES
63513e2a4803   nodered/node-red:latest             "/entrypoint.sh"        58 seconds ago Up 55 seconds (healthy)
0.0.0.0:1880->1880/tcp, [::]:1880->1880/tcp
nodered
38bc49be18a4   emqx/emqx:latest                    "/usr/bin/docker-ent..." About a minute ago Up About a minute
4370/tcp, 5369/tcp, 8083-8084/tcp, 0.0.0.0:1883->1883/tcp, [::]:1883->1883/tcp, 0.0.0.0:18083->18083/tcp, [::]:18083->18083/tcp, 8883/tcp emqx
sofwansabri@iot-server:~$ http://<YOUR_EXTERNAL_IP>:18083
-bash: YOUR_EXTERNAL_IP: No such file or directory
sofwansabri@iot-server:~$
```

3.4 MQTT communication established between ESP32(postman) and EMQX broker.





3.5 Node-RED flow created to validate and forward data to MongoDB Atlas.



3.6 MongoDB Charts configured for real-time dashboard visualization.

